

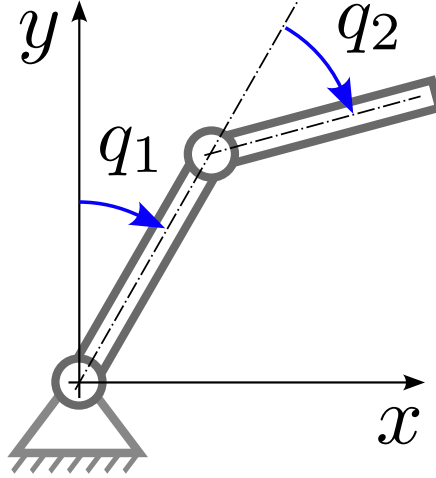


Figure 1: The Pendubot system.

1 Example: Periodic Motion Planning for the Pendubot System

To illustrate the main contribution of this paper, we explore the existence of forced periodic trajectories of the Pendubot system [2]. The system represents a two links robot with revolute joints that moves within a vertical plane. The first joint is actuated by a DC motor, while the second joint rotates freely. Let the generalized coordinates $q \in \mathbb{R}^2$ be set as in Figure 1 and let the DC motor operate in torque control mode. Then, dynamics of the robot are written as

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = Bu \quad (1)$$

with matrix coefficients

$$\begin{aligned} M(q) &= \begin{pmatrix} p_1 + 2p_2 \cos q_2 & p_3 + p_2 \cos q_2 \\ p_3 + p_2 \cos q_2 & p_3 \end{pmatrix}, \\ C(q, \dot{q})\dot{q} &= p_2 \sin q_2 \begin{pmatrix} -2\dot{q}_2\dot{q}_1 - \dot{q}_2^2 \\ \dot{q}_1^2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \\ G(q) &= g \begin{pmatrix} -p_4 \sin q_1 - p_5 \sin(q_1 + q_2) \\ -p_5 \sin(q_1 + q_2) \end{pmatrix}, \end{aligned}$$

where $g, p_1, \dots, p_5$ are physical parameters of the robot, depending on geometric relations and mass distribution. We set

$$\begin{aligned} p_1 &= 0.4 \text{kg} \cdot \text{m}^2, & p_2 &= 0.1 \text{kg} \cdot \text{m}^2, & p_3 &= 0.1 \text{kg} \cdot \text{m}^2, \\ p_4 &= 0.3 \text{kg} \cdot \text{m}, & p_5 &= 0.1 \text{kg} \cdot \text{m}, & g &= 9.81 \frac{\text{m}}{\text{s}^2}. \end{aligned}$$

For the Pendubot, we propose identifying configuration space points, a small neighborhood of which possesses periodic trajectories. If we limit ourselves to the case of regular reduced dynamics, then to solve this problem we may consider a linear servo constraint

$$\Phi(\theta) = q_e + v\theta$$

with constants $v \in \mathbb{R}^2$, $q_e \in \mathbb{R}^2$. According to Theorem 3 formulated in [1] we require that the resulting reduced dynamics exhibit a center-type equilibrium at the point q_e . This is achieved when q_e satisfies the equation

$$\gamma(0) = B_{\perp} G(q_e) = 0 \quad (2)$$

and there exists v , such that the system of inequalities

$$\alpha(0) = B_{\perp} M(q_e) v \neq 0 \quad \text{and} \quad (3)$$

$$\frac{\gamma'(0)}{\alpha(0)} = \frac{B_{\perp} \left(\frac{\partial G}{\partial q} \right)_{q=q_e} v}{B_{\perp} M(q_e) v} > 0$$

holds true. The system (2, 3) is written explicitly as

$$\begin{aligned} \sin(q_{e,1} + q_{e,2}) &= 0, \\ \cos(q_{e,1} + q_{e,2}) v_1 + \cos(q_{e,1} + q_{e,2}) v_2 &< 0, \\ (p_3 + p_2 \cos q_{e,2}) v_1 + p_3 v_2 &> 0 \end{aligned}$$

and it is solvable when

$$q_{e,1} + q_{e,2} = \pi k, \quad q_{e,1} \neq \frac{\pi}{2} + \pi n, \quad k, n \in \mathbb{Z}.$$

As a result, we conclude that the Pendubot possesses periodic trajectories in a vicinity of configurations where the second link of the robot is oriented vertically, while the first link can take any position except for the horizontal one.

The question immediately arises: *are there periodic trajectories near other configurations of the robot? For example, does a periodic trajectory exist in which the robot's second link oscillates near the horizontal position?*

To address this question, let us consider the case when the reduced dynamics are not supposed to be regular. In this regard, we introduce the servo constraint

$$\Phi(\theta) = q_s + \begin{pmatrix} -p_3 \\ p_3 + p_2 \cos q_{2,s} \end{pmatrix} \theta + \frac{k}{2p_3} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \theta^2, \quad (4)$$

where $k \in \mathbb{R}$, $q_s \in \mathbb{R}^2$ are constant parameters. If we select $B_{\perp} = (0, 1)$, then the coefficients α, β, γ of the reduced dynamics

$$\alpha(\theta) \ddot{\theta} + \beta(\theta) \dot{\theta}^2 + \gamma(\theta) = 0$$

evaluate to

$$\begin{aligned} \alpha(\theta) &= k\theta + p_2 p_3 \cos q_{s,2} - p_2 p_3 \cos \left(q_{s,2} + p_3 \theta + p_2 \theta \cos q_{s,2} + \frac{k\theta^2}{2p_3} \right), \\ \beta(\theta) &= k + p_2 p_3^2 \sin \left(p_3 \theta + p_2 \theta \cos q_{s,2} + q_{s,2} + \frac{k\theta^2}{2p_3} \right), \\ \gamma(\theta) &= -g p_5 \sin \left(q_{s,1} + q_{s,2} + p_2 \theta \cos q_{s,2} + \frac{k\theta^2}{2p_3} \right). \end{aligned}$$

As is readily apparent that the constraint $\Phi(\theta)$ ensures the reduced dynamics become singular at $\theta = 0$, corresponding the robot's configuration q_s . To determine whether the singular point is permeable, we outline

the conditions of Theorem 1 for its neighborhood:

$$\begin{aligned} \alpha'(0) &= k + \frac{1}{2}p_2^2p_3 \sin 2q_{s,2} + p_2p_3^2 \sin q_{s,2} > 0 \quad \text{and} \\ \frac{\beta(0)}{\alpha'(0)} &= \frac{k + p_2p_3^2 \sin q_{s,2}}{k + \frac{1}{2}p_2^2p_3 \sin 2q_{s,2} + p_2p_3^2 \sin q_{s,2}} < -\frac{1}{2} \quad \text{and} \\ \gamma(0) &= -gp_5 \sin(q_{s,1} + q_{s,2}) > 0. \end{aligned} \tag{5}$$

We represent the conditions 5 as a system of linear inequalities in terms of k :

$$\begin{aligned} -p_2p_3^2 \sin q_{s,2} - \frac{1}{2}p_2^2p_3 \sin 2q_{s,2} &< k \\ -p_2p_3^2 \sin q_{s,2} - \frac{1}{6}p_2^2p_3 \sin 2q_{s,2} &> k \\ \sin(q_{s,1} + q_{s,2}) &< 0, \end{aligned}$$

which directly provides the solvability set: at a given point q_s there exists a k satisfying (5) iff $q_{s,2} \in (0, \frac{\pi}{2}) + \pi m$ and $q_{s,1} + q_{s,2} \in (-\pi, 0) + 2\pi n$, where $n, m \in \mathbb{Z}$.

It is easy to show that using a left annihilator of the opposite direction $B_\perp = (0, -1)$ alters the signs of the coefficients α, β, γ in the reduced dynamics. Consequently, the conditions of Theorem 1 lead to a different system of inequalities

$$\begin{aligned} -p_2p_3^2 \sin q_{s,2} - \frac{1}{2}p_2^2p_3 \sin 2q_{s,2} &> k \\ -p_2p_3^2 \sin q_{s,2} - \frac{1}{6}p_2^2p_3 \sin 2q_{s,2} &< k \\ \sin(q_{s,1} + q_{s,2}) &> 0. \end{aligned} \tag{6}$$

In this case the solvability criteria becomes: $q_{s,2} \in (-\frac{\pi}{2}, 0) + \pi m$ and $q_{s,1} + q_{s,2} \in (0, \pi) + 2\pi n$. As a result, we infer that the Pendubot in a vicinity of q_s satisfying

- either $q_{s,2} \in (0, \frac{\pi}{2}) + \pi k$ and $q_{s,1} + q_{s,2} \in (-\pi, 0) + 2\pi n$;
- or $q_{s,2} \in (-\frac{\pi}{2}, 0) + \pi k$ and $q_{s,1} + q_{s,2} \in (0, \pi) + 2\pi n$, where $n, k \in \mathbb{Z}$

possesses periodic trajectories. The set of such points is shown in Figure 2. As seen, there are configurations where the robot's second link is oriented horizontally. Specifically, if set $q_{s,1} = \frac{3\pi}{4}$ $q_{s,2} = -\frac{\pi}{4}$, $k = -p_2p_3^2 \sin q_{s,2} - \frac{1}{3}p_2^2p_3 \sin 2q_{s,2}$, then the conditions 6 are satisfied. Numerical integration of the corresponding reduced dynamics with initial conditions $\theta(0) = -0.02, \dot{\theta}(0) = 0$ and $\theta(\frac{T}{2}) = 0.02, \dot{\theta}(\frac{T}{2}) = 0$ yields the phase trajectory presented in Figure 3. We also reconstruct the phase trajectory $(q_*(t), \dot{q}_*(t))$ and control input $u_*(t)$ of the original system by substituting the numerically obtained trajectory $\theta_*(t)$ into the expressions in TODO. These plots are shown in Figure 3. As observed, all signals are smooth functions. It is important to note that this trajectory could not be obtained with a regular servo constraint. This indicates that the proposed approach uncovers a new class of feasible trajectories.

References

- [1] Shiriaev, A. and Robertsson, A. and Perram, J. and Sandberg, A., Periodic motion planning for virtually constrained Euler-Lagrange systems, *Systems & Control Letters*, volume 55, 2006.

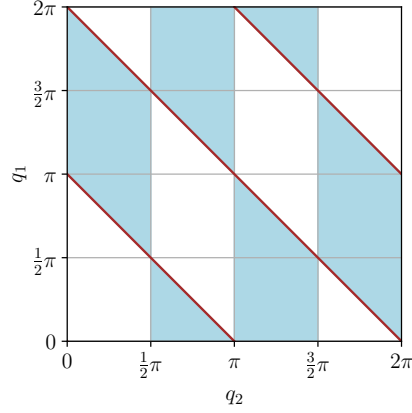


Figure 2: Configuration Space of the Pendubot. The blue regions indicate the configurations where the conditions of Theorem 1 may be satisfied with an appropriate servo constraint, and thus, periodic trajectories exist near each of these configurations. The brown line identifies the equilibrium points of the robot.

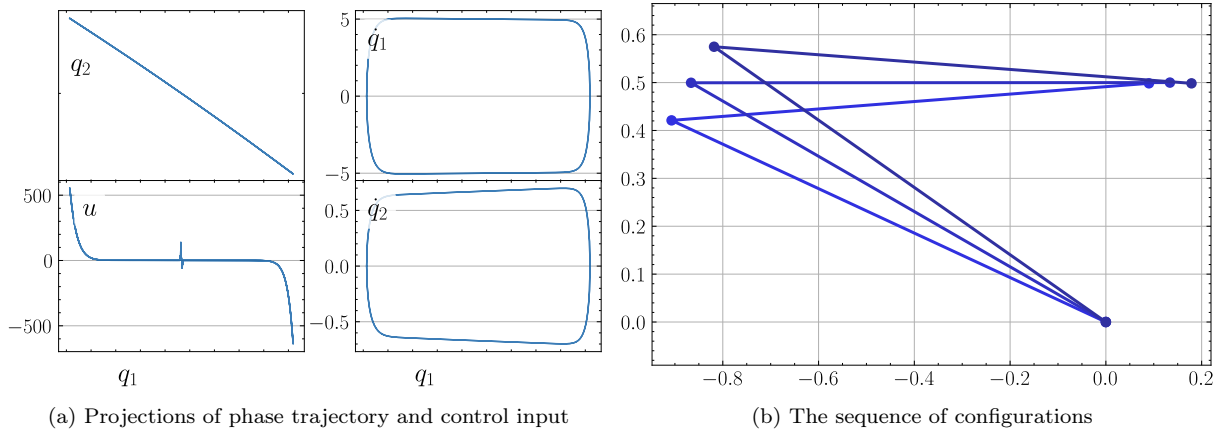


Figure 3: A periodic trajectory of the Pendubot system when its second links oscillates near horizontal position.

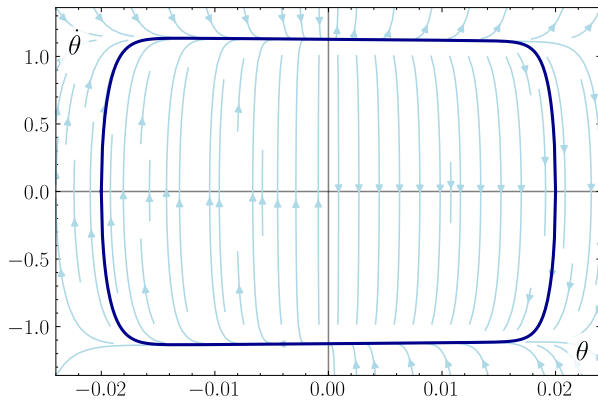


Figure 4: Phase portrait of the reduced dynamics of the Pendubot.

- [2] Spong, M.W., Underactuated Mechanical Systems, Control Problems in Robotics and Automation, B. Siciliano and K.P. Valavanis (Eds), Lecture Notes in Control and Information Sciences 230 Springer-Verlag, London, UK, 1997, presented at the International Workshop on Control Problems in Robotics and Automation: Future Directions Hyatt Regency, San Diego, California, Dec. 1997.